Compositional dynamics: ResourceSharers and Machines

Simon Frost

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Introduction

While Petri nets (see `vignette("composition")`) are an excellent formalism for mass-action kinetics, many dynamical systems don't fit neatly into the Petri net framework — especially when: - Rate laws are nonlinear or involve sums like total population N - The system includes **delay differential equations** (DDEs) - You want **directed** signal flow (inputs and outputs)

algebraicodin provides two additional formalisms, ported from AlgebraicDynamics.jl:

1. **ResourceSharers** — open dynamical systems with symmetric ports, composed via undirected wiring diagrams (UWDs)
2. **Machines** — open systems with directed inputs/outputs, composed via directed wiring diagrams (DWDs)

```
library(algebraicodin)
library(catlab)
library(acsets)
#>
#> Attaching package: 'acsets'
#> The following object is masked from 'package:graphics':
#>
#>     arrows
#> The following object is masked from 'package:base':
#>
#>     objects
```

Part 1: ResourceSharers

A `ResourceSharer` represents an open dynamical system where some internal states are *exposed* at ports. When composed, states at the same port are identified and their dynamics are summed.

Building an SIR model from two ResourceSharers

Instead of writing the SIR ODE system monolithically, we decompose it into two processes that share state variables:

```
# Infection process: affects S and I
infection <- continuous_resource_sharer(
  nstates = 2L,
  dynamics = function(u, p, t) {
    beta <- p$beta
    foi <- beta * u[1] * u[2]
    c(-foi, foi)  # dS, dI from infection
  },
  state_names = c("S", "I")
```

```r
)

# Recovery process: affects I and R
recovery <- continuous_resource_sharer(
  nstates = 2L,
  dynamics = function(u, p, t) {
    gamma <- p$gamma
    c(-gamma * u[1], gamma * u[1])  # dI, dR from recovery
  },
  state_names = c("I", "R")
)
```

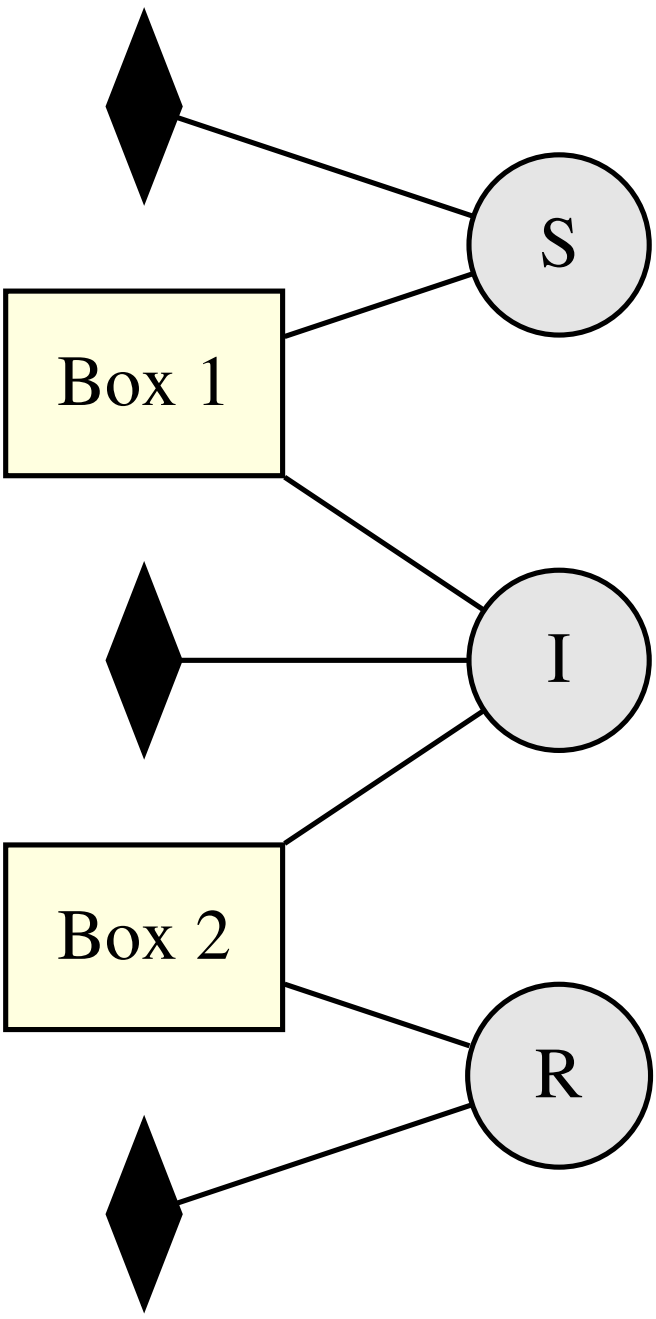
Each component has its own dynamics function with signature `(u, p, t)`. The `portmap` (default: identity) specifies which states are exposed.

Composing via a UWD

We compose these by specifying a wiring diagram that declares which ports (state variables) are shared:

```r
# Three junctions: S, I, R
# Infection uses S and I; Recovery uses I and R
sir_uwd <- uwd(c("S", "I", "R"),
  infection = c("S", "I"),
  recovery = c("I", "R")
)
```

```r
plot_uwd(sir_uwd)
```



Now compose:

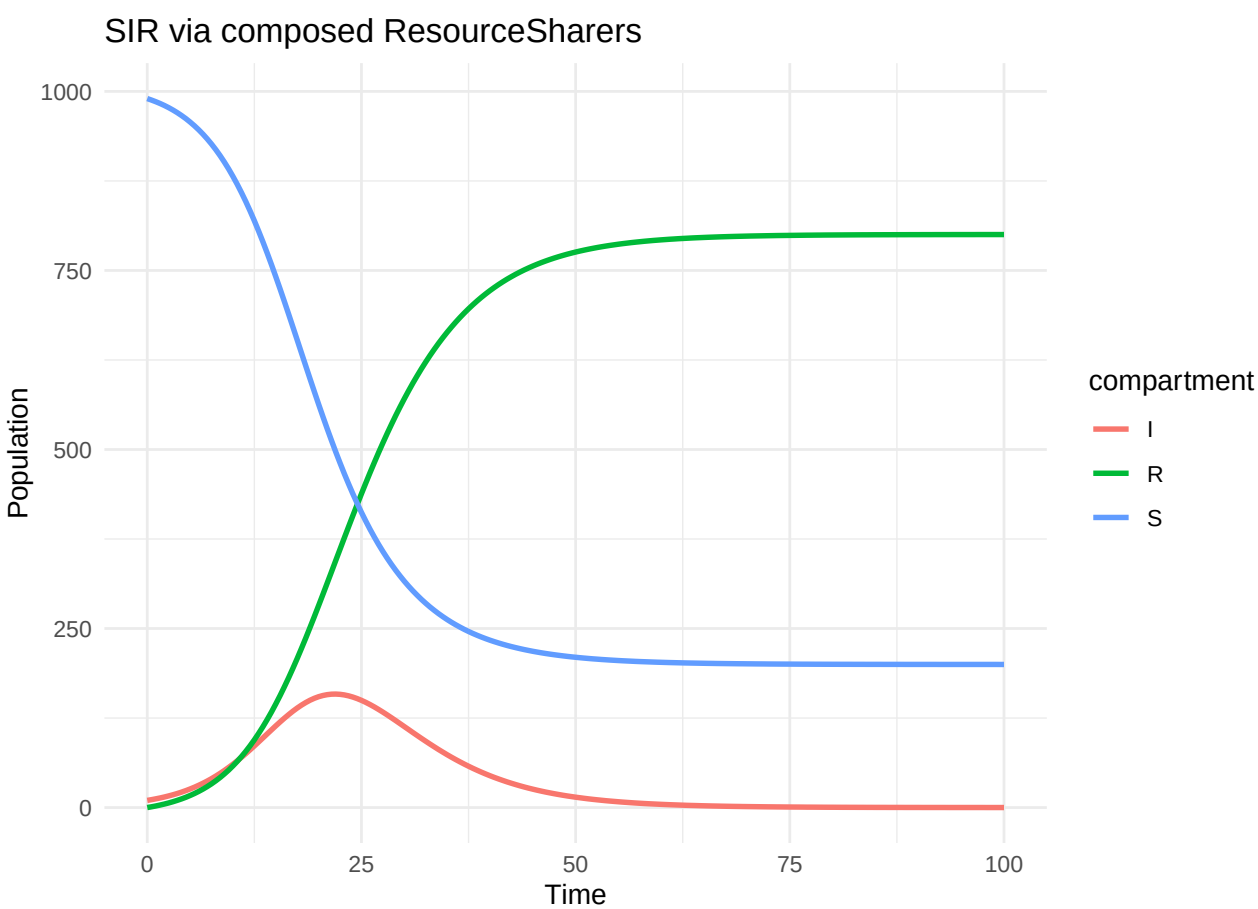
```
sir <- oapply_dynam(sir_uwd, list(infection, recovery))
sir@nstates
#> [1] 3
sir@state_names
#> [1] "S" "I" "R"
```

The composed system has 3 states — `I` is shared between the two processes, and its derivative is the *sum* of contributions from infection (+betaSI) and recovery (-gammaI).

Simulating with deSolve

```
result <- simulate_rs(sir,
  initial = c(S = 990, I = 10, R = 0),
  times = seq(0, 100, by = 0.5),
  params = list(beta = 0.4 / 1000, gamma = 0.2)
)

head(result)
#>   time        S        I        R
#> 1  0.0 990.0000 10.00000 0.000000
#> 2  0.5 987.9221 11.02737 1.050568
#> 3  1.0 985.6362 12.15499 2.208815
#> 4  1.5 983.1233 13.39150 3.485195
#> 5  2.0 980.3629 14.74601 4.891048
#> 6  2.5 977.3332 16.22815 6.438651
```

```
library(ggplot2)
df <- tidyr::pivot_longer(result, -time, names_to = "compartment",
                          values_to = "count")
ggplot(df, aes(x = time, y = count, color = compartment)) +
  geom_line(linewidth = 1) +
  labs(title = "SIR via composed ResourceSharers",
       x = "Time", y = "Population") +
  theme_minimal()
```



Comparing with a direct Petri net model

We can verify the composed ResourceSharer gives the same result as a Petri net:

```
sir_pn <- labelled_petri_net(
  c("S", "I", "R"),
  inf = c("S", "I") %=>% c("I", "I"),
  rec = "I" %=>% "R"
)
rs_petri <- petri_to_continuous(sir_pn)

# Evaluate at same state
u0 <- c(990, 10, 0)
p_composed <- list(beta = 0.4 / 1000, gamma = 0.2)
p_petri <- list(inf = 0.4 / 1000, rec = 0.2)

du_composed <- sir@dynamics(u0, p_composed, 0)
du_petri <- rs_petri@dynamics(u0, p_petri, 0)

cat("Composed:  ", du_composed, "\n")
#> Composed:   -3.96 1.96 2
cat("Petri net: ", du_petri, "\n")
#> Petri net:  -3.96 1.96 2
cat("Max diff:  ", max(abs(du_composed - du_petri)), "\n")
#> Max diff:   0
```

Lotka-Volterra: A non-epidemiological example

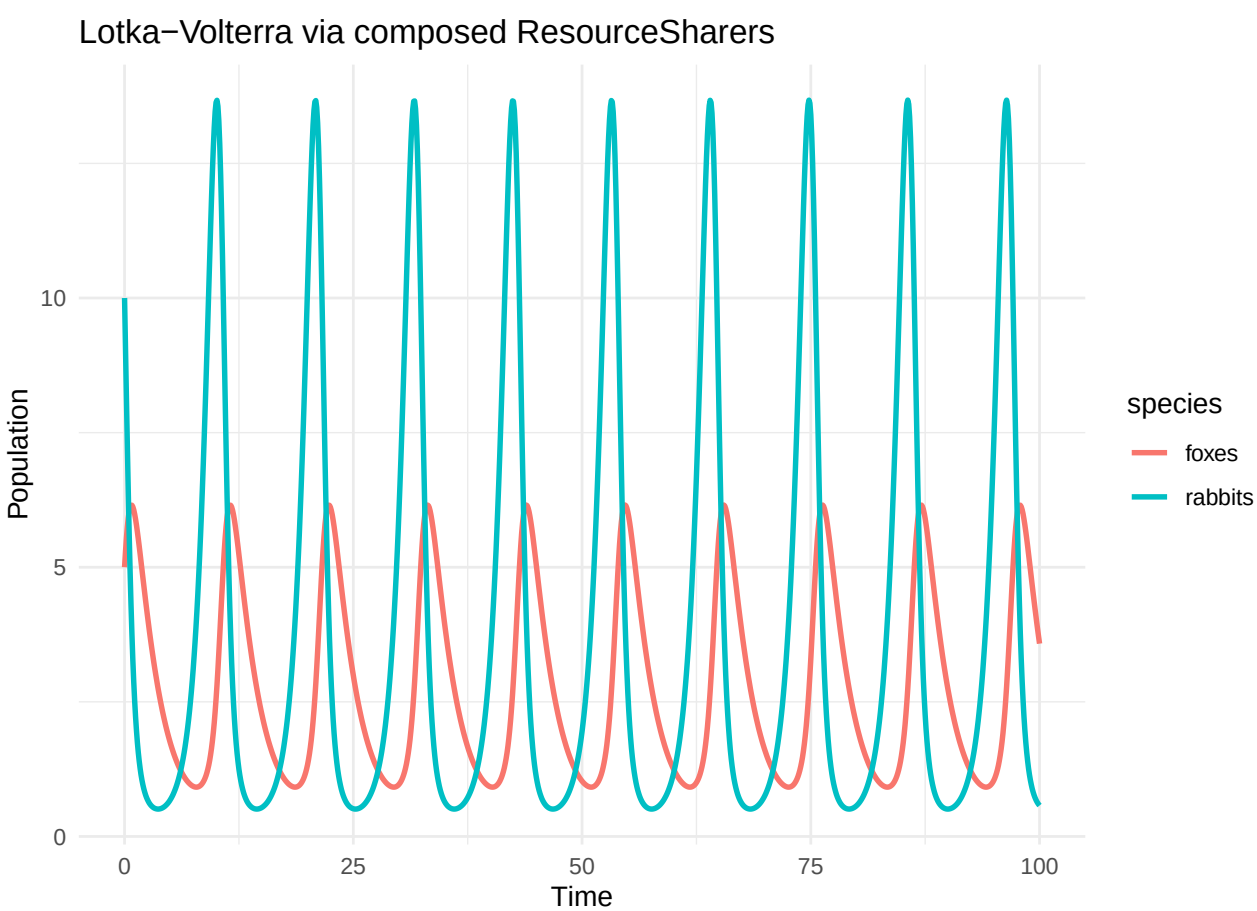
ResourceSharers are not limited to disease models. Here's the classic predator-prey system decomposed into three processes:

```
# Rabbit growth: dx/dt = alpha*x
rabbit_growth <- continuous_resource_sharer(
  nstates = 1L,
  dynamics = function(u, p, t) c(p$alpha * u[1]),
  state_names = c("rabbits")
)

# Fox death: dy/dt = -gamma*y
fox_death <- continuous_resource_sharer(
  nstates = 1L,
  dynamics = function(u, p, t) c(-p$gamma * u[1]),
  state_names = c("foxes")
)

# Predation: dx/dt = -beta*x*y, dy/dt = delta*x*y
predation <- continuous_resource_sharer(
  nstates = 2L,
  dynamics = function(u, p, t) {
    c(-p$beta * u[1] * u[2], p$delta * u[1] * u[2])
  },
  state_names = c("rabbits", "foxes")
)
```

```
lv_uwd <- uwd(c("rabbits", "foxes"),
  growth = c("rabbits"),
```

```r
  death = c("foxes"),
  predation = c("rabbits", "foxes")
)

lv <- oapply_dynam(lv_uwd, list(rabbit_growth, fox_death, predation))
lv@nstates
#> [1] 2
```

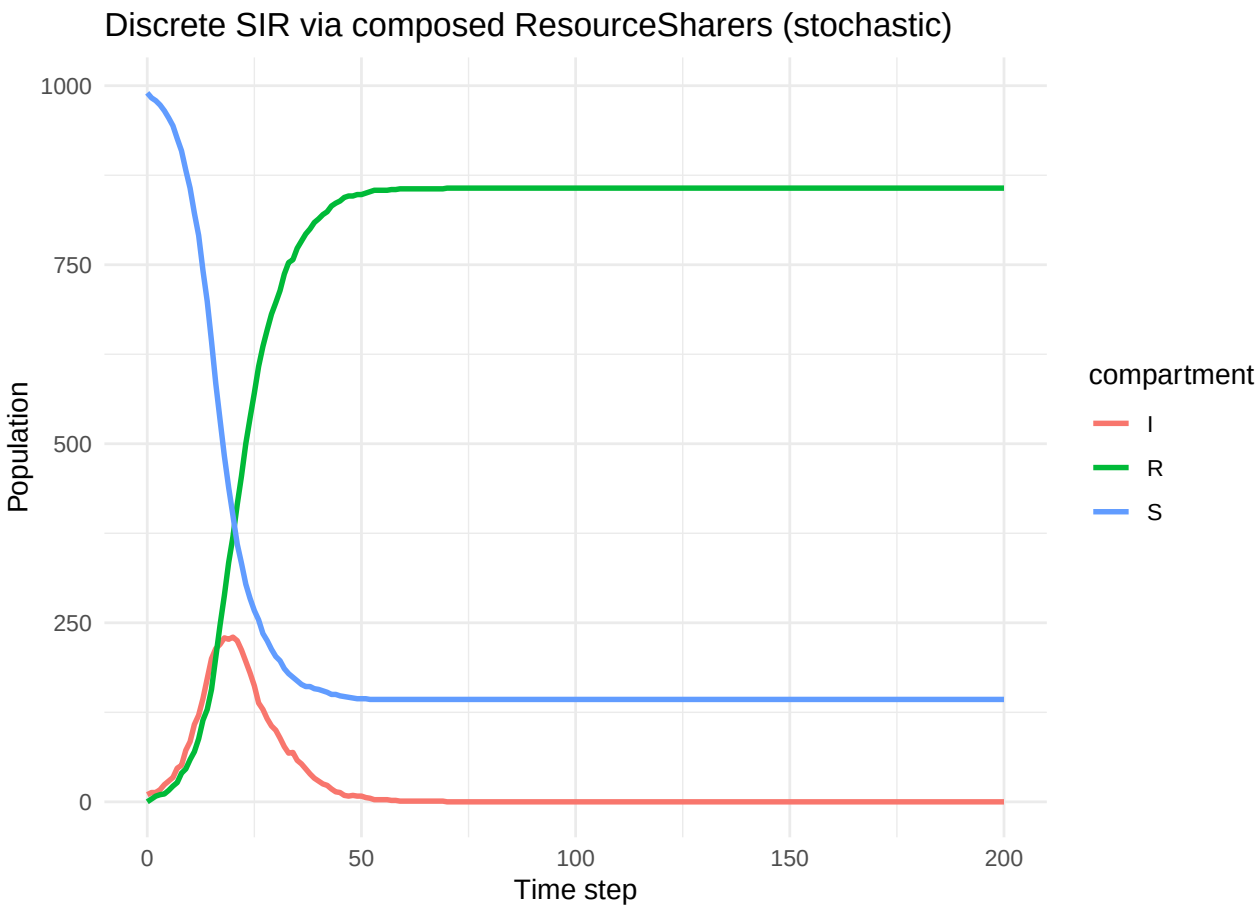
```r
lv_result <- simulate_rs(lv,
  initial = c(rabbits = 10, foxes = 5),
  times = seq(0, 100, by = 0.1),
  params = list(alpha = 1.1, beta = 0.4, gamma = 0.4, delta = 0.1)
)

df_lv <- tidyr::pivot_longer(lv_result, -time, names_to = "species",
                             values_to = "count")
ggplot(df_lv, aes(x = time, y = count, color = species)) +
  geom_line(linewidth = 1) +
  labs(title = "Lotka-Volterra via composed ResourceSharers",
       x = "Time", y = "Population") +
  theme_minimal()
```



Discrete ResourceSharers

The same composition framework supports discrete-time update rules:

```r
# Discrete SIR with step-wise updates
sir_discrete <- oapply_dynam(
  uwd(c("S", "I", "R"),
    infection = c("S", "I"),
    recovery = c("I", "R")
  ),
  list(
    discrete_resource_sharer(
      nstates = 2L,
      dynamics = function(u, p, t) {
        new_inf <- rbinom(1, round(u[1]), 1 - exp(-p$beta * u[2]))
        c(u[1] - new_inf, u[2] + new_inf)
      },
      state_names = c("S", "I")
    ),
    discrete_resource_sharer(
      nstates = 2L,
      dynamics = function(u, p, t) {
        new_rec <- rbinom(1, round(u[1]), 1 - exp(-p$gamma))
        c(u[1] - new_rec, u[2] + new_rec)
      },
      state_names = c("I", "R")
    )
  )
)

disc_result <- simulate_rs_discrete(sir_discrete,
  initial = c(S = 990, I = 10, R = 0),
  nsteps = 200,
  params = list(beta = 0.4 / 1000, gamma = 0.2)
)

df_disc <- tidyr::pivot_longer(disc_result, -time, names_to = "compartment",
                               values_to = "count")
ggplot(df_disc, aes(x = time, y = count, color = compartment)) +
  geom_line(linewidth = 1) +
  labs(title = "Discrete SIR via composed ResourceSharers (stochastic)",
       x = "Time step", y = "Population") +
  theme_minimal()
```



Part 2: Machines (Directed Composition)

A `Machine` has explicit **input** and **output** ports. Unlike ResourceSharers (which are symmetric), Machines have directed signal flow — the output of one machine feeds the input of another.

Building blocks

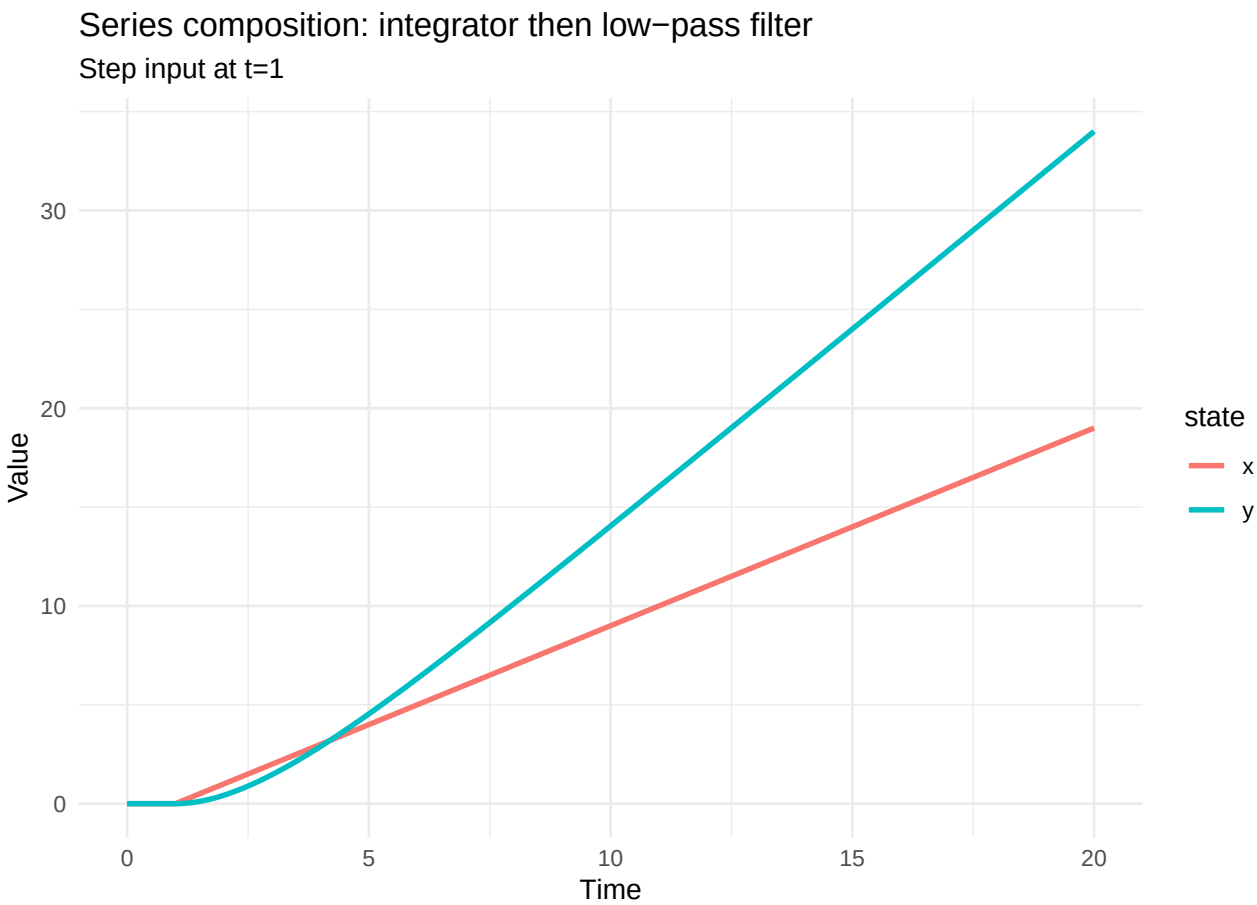
```
# An integrator: dx/dt = input, output = x
integrator <- continuous_machine(
  ninputs = 1L, nstates = 1L, noutputs = 1L,
  dynamics = function(u, x, p, t) x,        # dx/dt = input signal
  readout = function(u, p, t) u,             # output = state
  state_names = c("x")
)

# A filter: dx/dt = -alpha*x + input, output = x
lpf <- continuous_machine(
  ninputs = 1L, nstates = 1L, noutputs = 1L,
  dynamics = function(u, x, p, t) c(-p$alpha * u[1] + x[1]),
  readout = function(u, p, t) u,
  state_names = c("y")
)
```

Composing in series via a DWD

```r
# Wire: external_input -> integrator -> filter -> external_output
series_dwd <- dwd(
  ninputs = 1L, noutputs = 1L,
  boxes = list(c(1, 1), c(1, 1)),
  wires = list(
    c(0, 1, 1, 1),   # external input -> integrator
    c(1, 1, 2, 1),   # integrator output -> filter input
    c(2, 1, -1, 1)   # filter output -> external output
  )
)

series <- oapply_machine(series_dwd, list(integrator, lpf))
series@nstates
#> [1] 2
series@ninputs
#> [1] 1
series@noutputs
#> [1] 1
```

Simulating with external input

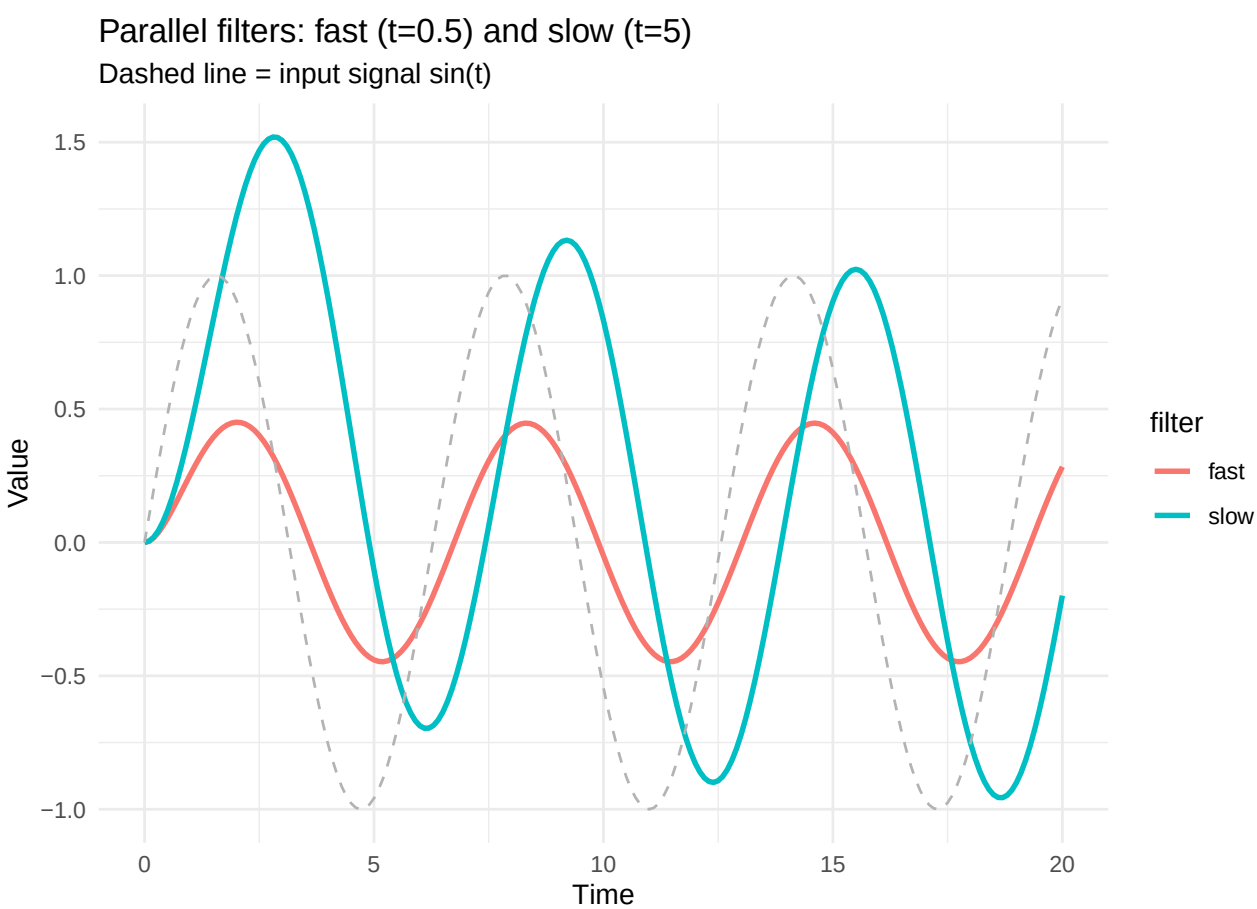
```r
# Drive with a step function input
result_m <- simulate_machine(series,
  initial = c(x = 0, y = 0),
  times = seq(0, 20, by = 0.1),
  input_fn = function(t) c(ifelse(t >= 1, 1, 0)),
  params = list(alpha = 0.5)
)

df_m <- tidyr::pivot_longer(result_m, -time, names_to = "state",
                            values_to = "value")
ggplot(df_m, aes(x = time, y = value, color = state)) +
  geom_line(linewidth = 1) +
  labs(title = "Series composition: integrator then low-pass filter",
       subtitle = "Step input at t=1",
       x = "Time", y = "Value") +
  theme_minimal()
```



Parallel machines

```
# Two independent filters in parallel
parallel_dwd <- dwd(
  ninputs = 2L, noutputs = 2L,
  boxes = list(c(1, 1), c(1, 1)),
  wires = list(
    c(0, 1, 1, 1),   # ext input 1 -> filter 1
    c(0, 2, 2, 1),   # ext input 2 -> filter 2
    c(1, 1, -1, 1),  # filter 1 -> ext output 1
    c(2, 1, -1, 2)   # filter 2 -> ext output 2
  )
)

fast_filter <- continuous_machine(
  ninputs = 1L, nstates = 1L, noutputs = 1L,
  dynamics = function(u, x, p, t) c(-2 * u[1] + x[1]),
  readout = function(u, p, t) u,
  state_names = c("fast")
)
slow_filter <- continuous_machine(
  ninputs = 1L, nstates = 1L, noutputs = 1L,
  dynamics = function(u, x, p, t) c(-0.2 * u[1] + x[1]),
  readout = function(u, p, t) u,
  state_names = c("slow")
```

```r
)

parallel <- oapply_machine(parallel_dwd, list(fast_filter, slow_filter))

result_p <- simulate_machine(parallel,
  initial = c(fast = 0, slow = 0),
  times = seq(0, 20, by = 0.1),
  input_fn = function(t) c(sin(t), sin(t)),
  params = list()
)

df_p <- tidyr::pivot_longer(result_p, -time, names_to = "filter",
                            values_to = "value")
ggplot(df_p, aes(x = time, y = value, color = filter)) +
  geom_line(linewidth = 1) +
  geom_line(aes(y = sin(time)), color = "grey70", linetype = "dashed") +
  labs(title = "Parallel filters: fast (t=0.5) and slow (t=5)",
       subtitle = "Dashed line = input signal sin(t)",
       x = "Time", y = "Value") +
  theme_minimal()
```



Euler approximation

Any continuous system can be converted to discrete-time via forward Euler:

```r
sir_continuous <- oapply_dynam(
  uwd(c("S", "I", "R"),
    infection = c("S", "I"),
    recovery = c("I", "R")
  ),
  list(infection, recovery)
)

sir_euler <- euler_approx(sir_continuous, h = 0.5)
sir_euler@system_type
#> [1] "discrete"

# One step
u0 <- c(990, 10, 0)
p <- list(beta = 0.4 / 1000, gamma = 0.2)
u1 <- sir_euler@dynamics(u0, p, 0)
cat("After one Euler step (h=0.5):", u1, "\n")
#> After one Euler step (h=0.5): 988.02 10.98 1
```

Converting Petri nets to dynamical systems

Any Petri net can be converted to a ResourceSharer for use with `oapply_dynam`:

```r
seir_pn <- labelled_petri_net(
  c("S", "E", "I", "R"),
  inf = c("S", "I") %=>% c("E", "I"),
  prog = "E" %=>% "I",
  rec = "I" %=>% "R"
)

# Continuous (ODE)
rs_ode <- petri_to_continuous(seir_pn)
cat("Continuous SEIR:", rs_ode@nstates, "states\n")
#> Continuous SEIR: 4 states

# Discrete (stochastic mass-action)
rs_disc <- petri_to_discrete(seir_pn)
cat("Discrete SEIR:", rs_disc@nstates, "states\n")
#> Discrete SEIR: 4 states

# Delay (DDE)
rs_dde <- petri_to_delay(seir_pn)
cat("Delay SEIR:", rs_dde@nstates, "states\n")
#> Delay SEIR: 4 states
```

Summary

Formalism	Composition	Use when…
Petri nets	`oapply()` via UWD	Mass-action kinetics, stoichiometric models
ResourceSharers	`oapply_dynam()` via UWD	General ODEs with shared state
Machines	`oapply_machine()` via DWD	Directed signal flow, control systems

All three can target odin2 for high-performance simulation, or be simulated directly via deSolve.